

Process Characterization Plan

A-Mab Drug Substance — Small-Virus Retentive Filtration (Step 9) — PCP-009

Process Development / Manufacturing Science & Technology (MSAT)

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Table of contents

Approvals	3
Abbreviations	3
1 Purpose and scope	5
2 Objectives	6
3 Related documents	7
4 Prior knowledge and quality risk basis	8
4.1 Unit-operation description and prior knowledge	8
4.2 Quality attributes in scope	8
4.3 Risk-based prioritization of parameters	9
5 Materials and methods	10
5.1 Scale-down model and its qualification	10
5.2 Operation	10
5.3 Analytical methods	10
5.4 Sampling plan	11
5.5 Statistical methods	11
6 Study design	12
6.1 Factors, ranges and study type	12
6.2 Screening design	12
6.3 Response-surface design	12
6.4 Univariate assessment	13
7 Acceptance and decision criteria	14
8 Data management and integrity	15
9 Roles and responsibilities	16
10 Deliverables and schedule	17
11 Risks and assumptions	18
12 Approvals	19
References	20
Appendix A — Planned screening design	21

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development / Virology	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

ANOVA, analysis of variance · CCD, central-composite design · CPP, critical process parameter · CQA, critical quality attribute · DoE, design of experiments · DS, drug substance · GPP, general process parameter · ICH, International Council for Harmonisation · KPP, key process parameter · LRF, log-reduction factor · LRV, log-reduction value · MSAT, Manufacturing Science and Technology · MVM, minute virus of mice (a parvovirus) · NOR, normal operating range · PAR, proven acceptable range · PPQ, process performance qualification · QbD, Quality by Design · RRF, risk ranking and filtering · RSM, response-surface methodology · SDM, scale-down model ·

TCID , 50 % tissue-culture infectious dose · WC-CPP, well-controlled CPP · XMuLV, xenotropic murine leukaemia virus (an enveloped retrovirus).

1 Purpose and scope

This Process Characterization Plan (**PCP-009**) defines the Stage 1 (Process Design) characterization of the A-Mab small-virus retentive filtration step (Step 9), the dedicated size-based virus-removal operation that follows the anion-exchange polish and precedes final ultrafiltration / diafiltration. Virus filtration removes small non-enveloped virus by a **mechanistic, size-exclusion mechanism that is orthogonal** to the low-pH inactivation (Step 6) and the charge-based anion-exchange clearance (Step 8); it is the **principal contributor to the cumulative MVM (parvovirus) log-reduction** and a major contributor to the enveloped-virus (XMuLV) log-reduction. The plan specifies the objectives, the attributes in scope, the process parameters and ranges to be characterized, the experimental designs, the analytical (virus-spiking) methods, the sampling plan, the statistical-analysis approach, and the acceptance and decision criteria by which each parameter will be classified and the volumetric-load limit established.

Scope. The commercial-scale small-virus retentive filtration step, characterized on a qualified scale-down filtration model with validated live-virus spiking studies. The upstream anion-exchange step (Step 8), which defines the feed to this step, and the downstream UF/DF step (Step 10) are addressed only where they define the feed to, or the feed from, this operation; their characterization is out of scope. This plan governs execution only — the results are reported in the paired Process Characterization Report (**PCR-009**).

Basis. The unit-operation model, quality-attribute framework and risk methodology follow the A-Mab case study (CMC Biotech Working Group 2009); viral clearance is evaluated and credited per ICH Q5A(R2) (International Council for Harmonisation 2023a), including orthogonal (modular) clearance and worst-case substantiation; the Stage 1 approach and report structure follow PDA TR 60 (Parenteral Drug Association 2013) and the ISPE Good Practice Guide (International Society for Pharmaceutical Engineering 2023); the QbD framework follows ICH Q8, Q9 and Q11 (International Council for Harmonisation 2009, 2023b, 2012) and the lifecycle approach of the FDA 2011 guidance (U.S. Food and Drug Administration 2011).

2 Objectives

The characterization will:

1. establish the quantitative relationship between the filtration parameters (volumetric load and pressure) and the MVM and XMuLV log-reduction and the step yield;
2. identify the significant main effects, interaction and curvature, including the anticipated decline of the MVM log-reduction with volumetric load;
3. establish the volumetric-load limit over which the credited small-virus log-reduction is preserved, and substantiate the credited clearance at the worst-case (maximum-load) condition per ICH Q5A(R2);
4. establish proven acceptable ranges (PARs) and confirm normal operating ranges (NORs) for each parameter;
5. classify each parameter for its risk of impact to the credited viral clearance and to process performance (CPP / WC-CPP / KPP / GPP); and
6. provide the process understanding and ranges that justify the Stage 2 operating conditions and the drug-substance viral-safety control strategy.

3 Related documents

This plan is executed under the master documents below and is paired with its report; the full package is cross-referenced for traceability.

Table 3.1: Related documents in the A-Mab process-characterization package.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-009	Process Characterization Report (Small-Virus Retentive Filtration (Step 9))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

The governing procedures and validated analytical methods referenced by this plan are listed in Table 3.2 (numbers are placeholders in this synthetic set).

Table 3.2: Referenced standard operating procedures and analytical method validations.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2012	Operation and Post-Use Integrity Testing of the Small-Virus Retentive Filtration Step	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	Method validation
AMV-3017	XMuvLV Infectivity Titre (TCID50)	Method validation

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

Small-virus retentive filtration is the dedicated virus-removal step of the A-Mab purification train: the anion-exchange flow-through pool is passed through a small-virus retentive (nanometre-rated) filter under controlled pressure, so that virus particles larger than the membrane rating are retained by size exclusion while the antibody monomer transmits (CMC Biotech Working Group 2009). Unlike the upstream operations it forms no new product-quality CQA; its role in the control strategy is to provide **robust, size-based removal of small non-enveloped virus (MVM)** together with substantial removal of larger enveloped virus (XMuLV), as an orthogonal contribution to viral safety. Platform knowledge established that removal is governed by size exclusion at the membrane rating: the larger enveloped XMuLV is retained essentially completely and robustly, whereas the small parvovirus MVM — whose size is closer to the membrane rating — is retained with a log-reduction that is high but declines as the membrane is progressively loaded, so the credited MVM clearance is a function of the volumetric load; and that filtration pressure, within a controlled range, does not materially change retention but must be controlled to preserve filter performance and integrity. This knowledge frames the attributes in scope and the prioritization of parameters for characterization.

4.2 Quality attributes in scope

Virus filtration introduces no CQA. It is a principal clearance step for the two model viruses whose cumulative, cross-step log-reduction is specified at the drug substance (Table 4.1): the enveloped retrovirus XMuLV and the small non-enveloped parvovirus MVM. Criticality is scored on the A-Mab continuum using Tool #1 (Risk Score = Impact $\times$ Uncertainty); both viral-clearance attributes are rated Very High (VH) and treated as *Critical* (CMC Biotech Working Group 2009; International Council for Harmonisation 2023a).

Table 4.1: Viral-clearance CQAs to which the virus-filtration step contributes (expressed as the cumulative, cross-step log-reduction), with acceptance criterion and criticality (Tool #1 = Impact $\times$ Uncertainty).

CQA	Category	Acceptance	Criticality	Tool #1
Viral clearance — MVM (parvovirus)	viral safety	8.6–20 log10 (cumulative)	VH	20
Viral clearance — XMuLV (enveloped)	viral safety	16.7–30 log10 (cumulative)	VH	20

The per-step MVM and XMuLV log-reductions are not themselves released specifications; they are the modular contributions whose sum, across the low-pH inactivation, anion-exchange and virus-filtration steps, must meet the cumulative log-reduction requirements at the drug substance. Because virus filtration is the principal MVM-removal mechanism, the study is designed to demonstrate the MVM-clearance capacity of the step at and beyond the maximum routine load; the MVM log-reduction is the primary response of the study.

4.3 Risk-based prioritization of parameters

Consistent with the A-Mab lifecycle of iterative risk assessments and the risk flow of PDA TR 60, the study scope is set by the Pre-Characterization Process Risk Assessment (**RA-001**). A risk-ranking-and-filtering (RRF) screen scores each parameter for its potential impact and assigns the study type accordingly: the volumetric load and the filtration pressure — which carry a credible risk to the credited viral log-reduction or must be controlled to preserve filter performance — are both studied in the multivariate DoE; no parameter at this step is assigned to univariate assessment (CMC Biotech Working Group 2009). The resulting parameter set and study-type assignment are given in Table 6.1.

5 Materials and methods

5.1 Scale-down model and its qualification

Characterization will be performed on a scale-down virus-filtration model operated at the commercial-process equivalents for membrane type and rating, area-normalized load, filtration pressure, buffer matrix and product concentration (**SOP-2012**). Prior to characterization the SDM will be qualified against at-scale reference data (**SOP-1001**) by comparing the operational parameters and, critically, the input and output attributes — volumetric throughput, MVM and XMuLV log-reduction and step yield — to confirm the model is representative and suitable to support the load limit and PAR claims (Parenteral Drug Association 2013; International Council for Harmonisation 2023a). Viral log-reduction will be determined by validated live-virus spiking studies on the qualified SDM. The centre-point replicates of each design provide an in-study estimate of model reproducibility (pure error) used in the lack-of-fit assessment.

5.2 Operation

The anion-exchange flow-through pool is filtered through a small-virus retentive membrane of the platform type and rating under controlled filtration pressure, to a defined area-normalized volumetric load, with the filtrate collected as the virus-filtration pool. Product transmits through the membrane while virus larger than the membrane rating is retained by size exclusion. Membrane lot, pre-use flush and buffer matrix are controlled to platform specifications and treated as identity/quality-controlled inputs rather than characterized variables (CMC Biotech Working Group 2009). A post-use filter-integrity test (**SOP-2012**) confirms membrane integrity after each run and is a release-linked control for the credited retention.

5.3 Analytical methods

The step responses will be measured by the validated methods in Table 5.1; method performance characteristics are per the cited validation reports. The MVM and XMuLV log-reduction are derived from the infectious titre of the filter load and filtrate in validated live-virus spiking studies.

Table 5.1: Validated analytical methods for the characterization.

Reference	Title	Type
AMV-3018	MVM Infectivity Titre (TCID ₅₀ /qPCR)	Method validation
AMV-3017	XMuLV Infectivity Titre (TCID ₅₀)	Method validation

5.4 Sampling plan

Each DoE run will be sampled at the filter load and the virus-filtration pool for the MVM and XMuLV infectious titre (spiking studies) and for the step-yield determination; in-process readings (pressure, flux/throughput and the post-use integrity result) will confirm the filtration profile and membrane integrity. Centre-point runs are replicated within each design (three in screening, four in the response-surface design) to quantify reproducibility and pure error.

5.5 Statistical methods

Designs will be constructed and analyzed in coded factors, with each parameter's characterization range mapped so that the set-point is 0 and the range edges are ± 1 (**SOP-1002**). Screening data will be analyzed by ordinary-least-squares regression of each response on the main effects and the two-factor interaction; factor effects are reported as twice the coded coefficient. Response-surface data will be analyzed with the full quadratic model. Significance will be assessed at $\alpha = 0.05$. Model adequacy will be judged by R^2 , adjusted R^2 , predicted R^2 (from the PRESS statistic), the overall model F-test, and an ANOVA lack-of-fit test against the centre-point pure error; a response with no significant parameter effect will be reported as robust rather than modelled predictively. The volumetric-load limit will be established from the MVM response and the credited clearance substantiated conservatively at the worst-case (maximum-load) condition per ICH Q5A(R2). Commercial-scale capability of the cumulative viral-clearance CQAs to which the step contributes will be estimated by Monte-Carlo simulation with each parameter varied within its NOR.

6 Study design

6.1 Factors, ranges and study type

Two parameters are in scope (Table 6.1). The characterization range of each factor (“Range studied”) defines the edges of the explored knowledge space and the basis of the PAR; the NOR is the tighter routine-control range. Classification is an output of the study and is reported in PCR-009. The DoE factor coding is given in Table 6.2.

Table 6.1: Virus-filtration parameters, set-points, characterization ranges and planned study type.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Filtration volume (load)	L/m2	90	50–140	50–105	multivariate
Filtration pressure	psi	13	8–30	10–20	multivariate

Table 6.2: Design-of-experiments factor legend (coded A–B).

Code	Factor	Unit	Studied in
A	Filtration volume (load)	L/m2	screening + RSM
B	Filtration pressure	psi	screening + RSM

6.2 Screening design

Factor screening will use a two-level full factorial in the two factors (A = volumetric load, B = filtration pressure) with three centre-point replicates, 7 runs in total. The full factorial estimates both main effects and the two-factor interaction without confounding. Three responses will be recorded for each run (MVM log-reduction, XMuLV log-reduction and step yield). The planned coded design is given in Appendix A.

6.3 Response-surface design

The two factors will be carried into a face-centred central-composite design with four centre-point replicates, 12 runs in total, enabling estimation of curvature in addition to the main effects and interaction. The planned coded design is given in Appendix B.

6.4 Univariate assessment

No parameter at this step is assigned to univariate assessment: both the volumetric load and the filtration pressure are carried into the multivariate design, the load because it governs the credited MVM clearance and the pressure so that its (expected small) effect on retention is quantified jointly with the load rather than assumed.

7 Acceptance and decision criteria

The study is acceptable when:

1. the SDM qualification confirms representativeness of commercial scale (**SOP-1001**) and the spiking studies meet their validity criteria (ICH Q5A(R2));
2. the fitted response-surface model for the MVM log-reduction is adequate — no significant lack of fit ($p > 0.05$ against the centre-point pure error) and residual diagnostics without systematic structure — and resolves the volumetric-load dependence (a response with no significant factor effect is reported as robust); and
3. a volumetric-load limit exists at or below which the credited MVM log-reduction is preserved with margin at the worst-case condition, with the XMuLV log-reduction and step yield robust across the NORs.

Each parameter will then be classified per the A-Mab continuum (CMC Biotech Working Group 2009):

- a parameter whose variability significantly impacts the credited viral clearance is a **CPP**; where the impact is reliably controlled within the operating region the parameter is a **well-controlled CPP (WC-CPP)**;
- a parameter that does not significantly impact the credited clearance but affects process performance or is controlled to protect it is a **KPP** or **WC-CPP** as appropriate; otherwise it is a **GPP**.

The load limit, classifications and PARs established here will justify the Stage 2 operating ranges and feed the drug-substance viral-safety control strategy.

8 Data management and integrity

All raw data, DoE designs, spiking-study results and statistical analyses will be recorded and retained under the site data-integrity procedures (ALCOA+). Analyses will be performed with version-controlled scripts and reviewed by a second analyst; designs, models and the derived load limit will be archived with the report. Any deviation from this plan will be documented and assessed for impact in PCR-009.

9 Roles and responsibilities

Table 9.1: Roles and responsibilities.

Function	Responsibility
Process Development / MSAT	Author the plan; design, execute and analyze the DoE; author PCR-009
Analytical Development / Virology	Perform the validated virus-spiking (infectivity) assays
Manufacturing Science	Qualify the scale-down filtration model; confirm commercial-scale representativeness
Statistics	Review the design, the statistical analysis and the load-limit definition
Quality Assurance	Approve the plan and the report; confirm parameter classifications

10 Deliverables and schedule

The deliverable of this plan is the Process Characterization Report (**PCR-009**), reporting the executed designs, the significant effects, the volumetric-load limit, the parameter classification and the viral-clearance outcomes. PCR-009 rolls up into the Process Characterization Master Report (**PCMR-001**) and provides the parameter risk basis for the post-characterization process risk assessment. Execution follows SDM qualification and precedes the Stage 2 PPQ campaign.

11 Risks and assumptions

The plan assumes the SDM is representative of commercial scale (confirmed by qualification) and that the spiking assays are validated and in control. The principal risk is that the absolute log-reduction or throughput shifts at commercial scale or across membrane lots, or that the load presented to the filter is higher than characterized; this is mitigated by defining the load limit on the qualified SDM, substantiating the credited clearance at the worst-case load, confirming the limit at scale in Stage 2, and carrying the WC-CPP ranges forward with the load limit, pressure control and the post-use filter-integrity test.

12 Approvals

Approval of this plan authorizes execution of the characterization described herein. Signatures are recorded in Table [1](#).

References

- CMC Biotech Working Group. 2009. *A-Mab: A Case Study in Bioprocess Development*. CASSS.
- International Council for Harmonisation. 2009. *ICH Q8(R2): Pharmaceutical Development*. ICH.
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- International Society for Pharmaceutical Engineering. 2023. *Good Practice Guide: Practical Implementation of the Lifecycle Approach to Process Validation*. ISPE.
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- U.S. Food and Drug Administration. 2011. *Guidance for Industry: Process Validation — General Principles and Practices*. FDA.

Appendix A — Planned screening design

Coded two-level full-factorial design (factors A–B per Table 6.2; coded levels $-1/0/+1$). Responses will be recorded at execution and reported in PCR-009.

Table 12.1: Appendix A — planned screening design (coded factor settings).

Run	Type	A	B
1	factorial	-1	-1
2	factorial	-1	1
3	factorial	1	-1
4	factorial	1	1
5	center	0	0
6	center	0	0
7	center	0	0

Appendix B — Planned response-surface design

Coded face-centred central-composite design (factors A–B; coded levels $-1/0/+1$).

Table 12.2: Appendix B — planned response-surface design (coded factor settings).

Run	Type	A	B
1	factorial	-1	-1
2	factorial	-1	1
3	factorial	1	-1
4	factorial	1	1
5	axial	-1	0
6	axial	1	0
7	axial	0	-1
8	axial	0	1
9	center	0	0
10	center	0	0
11	center	0	0
12	center	0	0